Typed contracts and schemas

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask which gate failed and what passing still does not establish.

Use exact-type examples, clear range failures and exposure-aware assessment rather than yes/no guessing.

Keep one habit: Model validation → diagnose type/value confusion → inspect probe outcomes → collect individual explanations → assess a fresh record later.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of typed contracts and **schema**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Architecture and data flow. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of architecture and data flow

Reference example, not the only acceptable wording: Teacher explanation: Architecture assigns responsibilities to components and shows information crossing trust boundaries. A diagram should include failure paths.

Trace request validator -> role filter -> source lookup -> response validator -> result. Sources enter as data, not control instructions. Reject invalid requests before any downstream lookup.

Pictured end goal

Explain typed contracts and **schema**s using the deciding step in this fictional service.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Architecture and data flow

Use the three-session teaching cycle above, with extra time as needed. Typed contracts define values, required fields and failure outcomes. Runtime **input** still needs validation even when program annotations exist. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of typed contracts and **schema**s
</details>

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach structure, type, range and domain as distinct checks

> Keep room requests separate from pictured sample records.

Definitions and complete closed-object contract

Typed contracts define values, required fields and failure outcomes. Runtime **input** still needs validation even when program annotations exist.

Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool. Reject extra fields, missing fields and wrong types before **process**ing.

Contract part	Exact rule in this exercise
Whole request	A Python dict with exactly room_id and requested_count
room_id	A nonempty Python string; no room existence or format check yet
requested_count	Exact Python int type and value at least zero; bool excluded
Unexpected or missing fields	Reject the request before processing
Type conversion	None: do not turn quoted digits or floats into integers
Failure outcome	ValueError with the relevant field/type/range category

Contract part	Exact rule in this exercise
Valid output	A copied validated dictionary, not a booking or permission grant

Ordered gates and main Boolean reasoning

Validate the outer structure before individual fields

GATE	QUESTION AND OUTCOME
1. Object and exact keys	Is this exactly a dict with both required fields and no extras? Otherwise reject.
2. Room identifier	Is room_id a nonempty string? Otherwise reject.
3. Count type	Is type(requested_count) exactly int? Reject bool, strings and other types.
4. Count range	Is the integer at least zero? Otherwise reject.
5. Return validated copy	Proceed to later authorization and domain checks; none has happened yet.

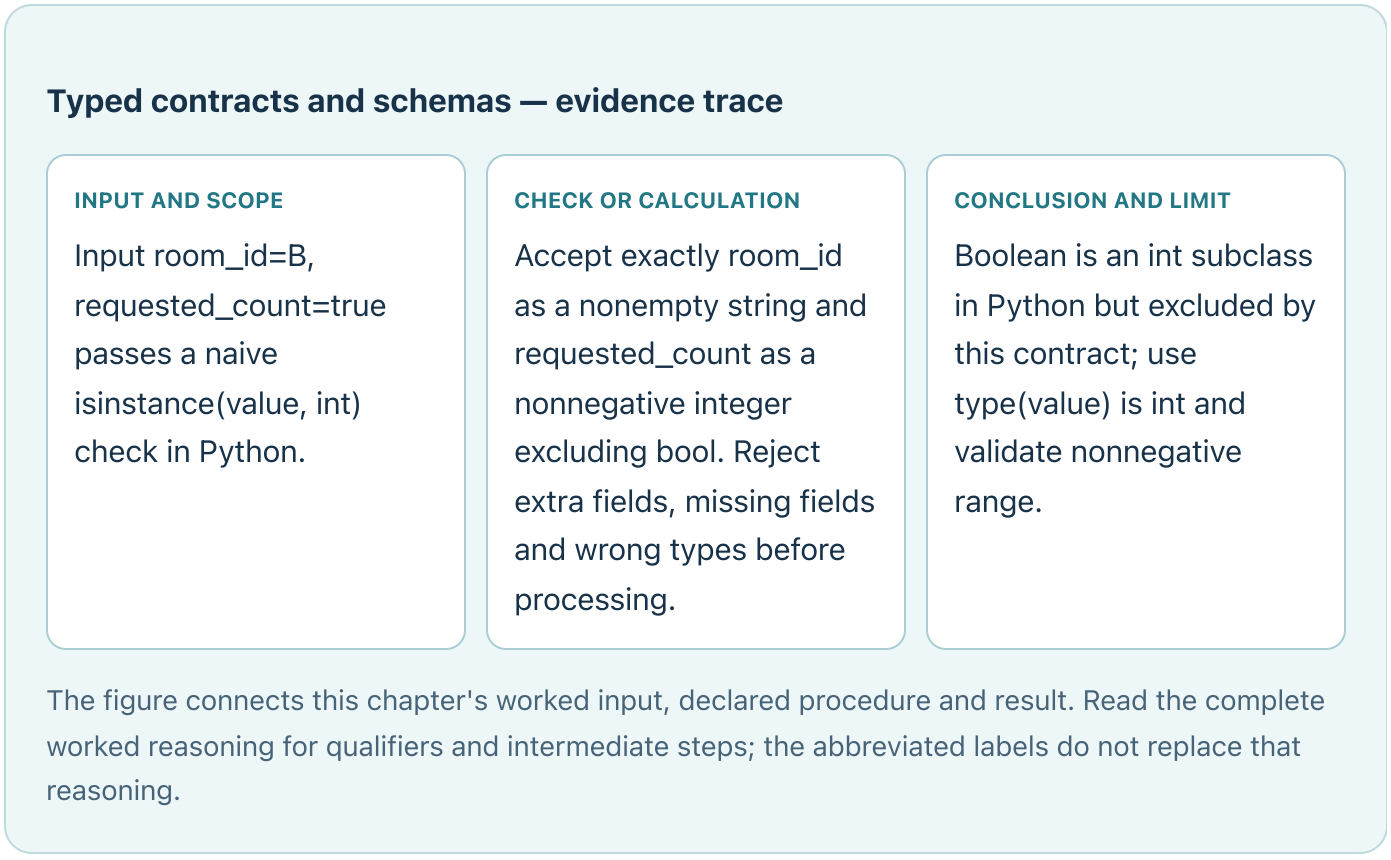
Passing the shape check does not imply that a room exists, the count is accurate or the caller may act.

Python has a subtle detail: isinstance(True, int) is True because bool is an int subclass. Our contract deliberately excludes Booleans, so type(True) is int is False. For the main record {room_id: "B", requested_count: True}, the two required fields and room string pass, but the count fails the exact-type gate. Do not silently reinterpret True as a request for one place.

Python True corresponds to JSON true when decoding JSON.

Require exact field names, exact integer type excluding bool, and nonnegative range. Distinguish invalid structure, wrong type and wrong value. The schema only requires a nonempty room-ID string; whitespace passes that literal check, with domain validation still needed. Original count 0 has already been modelled, so score that component as recall rather than entirely fresh transfer.

Collect individual reasons for the remaining count types and use a fresh record-level case for later independent assessment; do not silently award independence to a previously revealed answer.



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Boolean is an int subclass in Python but excluded by this contract; use type(value) is int and validate nonnegative range.

Qualification: **Schema**-valid **input** can still be unauthorised or semantically unsuitable; subsequent checks remain necessary.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Input room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python.

Boolean is an int subclass in Python but excluded by this contract; use type(value) is int and validate nonnegative range.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Domain limits and additional checks

What passes or fails?	What it establishes and what remains
Correct fields, room B, integer count 2	Schema-valid only; room existence/permission/capacity not checked
Unexpected book field	Reject extra field; do not ignore it or execute it
Missing requested_count	Reject incomplete structure; do not invent a default
Empty room_id string	Reject room ID under the nonempty rule
Whitespace-only room_id	Nonempty string passes this literal schema; format/domain checks are separate
A valid-looking count	Does not establish truthful measurements
A later authorization rule	Must still run after structure validation before any permitted operation

Separate pictured sample schema and scope

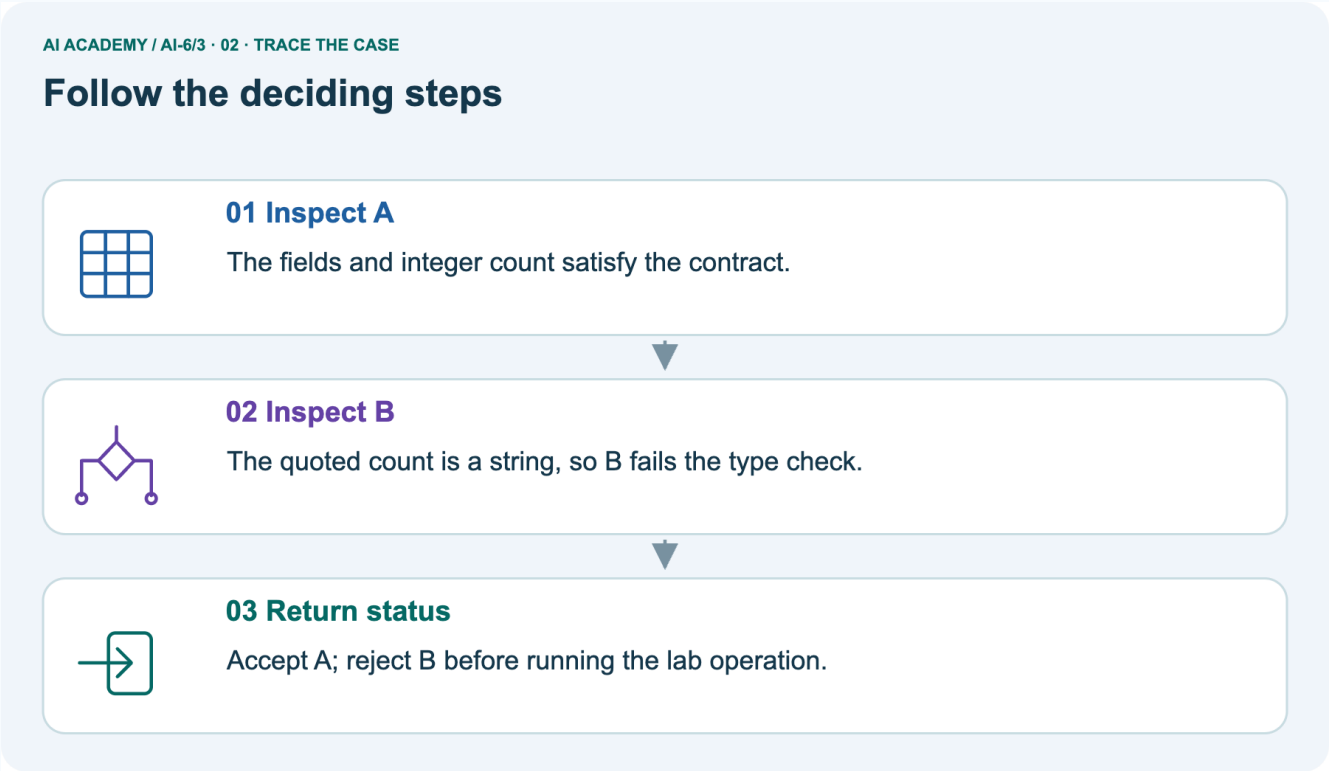
A labelled sample record. A lab tool accepts a string sample_id and an integer count of at least zero.

A={sample_id:"jar -8",count:4}; B={sample_id:"jar -9",count:"4"}. No type conversion is allowed.

The main room schema uses room_id and requested_count. The picture is a separate lab-sample schema using sample_id and count, with no conversion allowed. A has sample_id "jar -8" and integer count 4; B has "jar -9" and string count "4". A satisfies the pictured structure/type rules; B fails the count type. Changing A's count to –1 keeps the integer type but fails the nonnegative value rule.

A structurally valid count is not evidence that anyone counted the sample correctly.

Picture trace and negative-count key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Inspect A	The fields and integer count satisfy the contract.
2. Inspect B	The quoted count is a string, so B fails the type check.
3. Return status	Accept A; reject B before running the lab operation.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **schema** checks declared structure. Looking like a number is different from having the required numeric type.

Valid structure does not establish that the reported count is accurate.

Change: Change A count to -1.

Prediction: Reject A on the non-negative value rule.

Reason: The type is correct but the value is outside its accepted range.

▼ Reference: original optional lab

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def nonnegative_integer(value):
    return type(value) is int and value >= 0

def validate_request(value):
    if type(value) is not dict or set(value) != {'room_id', 'requested_count'}:
        raise ValueError('request fields')
    if type(value['room_id']) is not str or not value['room_id']:
        raise ValueError('room ID')
    if not nonnegative_integer(value['requested_count']):
        raise ValueError('requested count')
    return dict(value)

print(validate_request({"room_id": "A", "requested_count": 0}))
```

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Schema	A contract for structure and types
Runtime validation	Checking actual incoming values
Closed object	An object rejecting unexpected fields

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

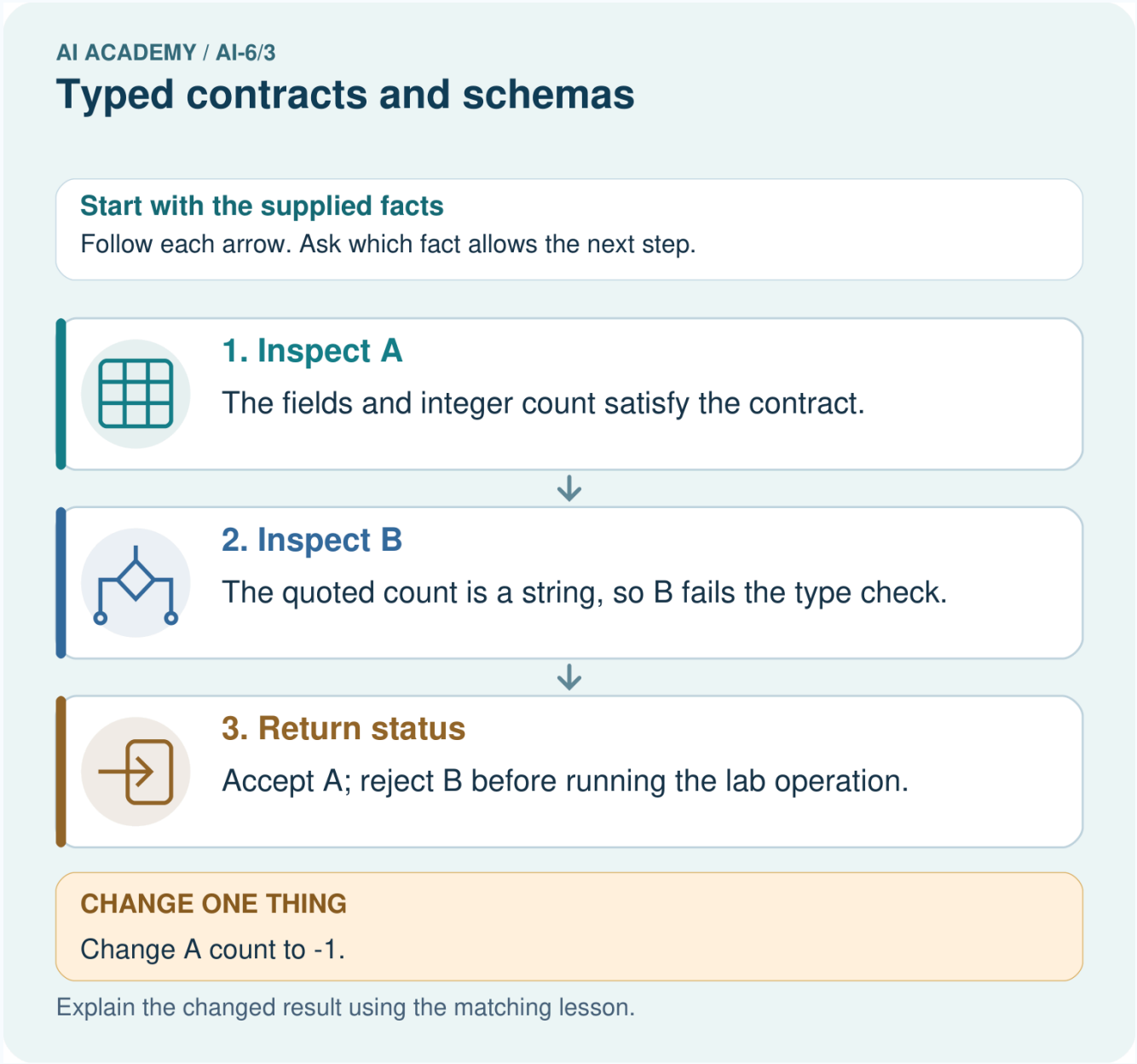
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Typed contracts and schemas

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: **Input** room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Boolean is an int subclass in Python but excluded by this contract; use type(value) is int and validate nonnegative range.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Typed contracts and **schema**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: **Schema**-valid **input** can still be unauthorised or semantically unsuitable; subsequent checks remain necessary.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Compare A's count 4 with B's count "4". Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?

The fields and integer count satisfy the contract. The quoted count is a string, so B fails the type check. Accept A; reject B before running the lab operation. Changed case: Reject A on the non-negative value rule.

The type is correct but the value is outside its accepted range.

Changed-case reasoning: Reject A on the non-negative value rule. The type is correct but the value is outside its accepted range.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main room B/True fails exact int type; do not coerce. Closed object rejects extra/missing fields before domain operations. Pictured integer 4 accepts, quoted 4 fails type, changed–1 fails range. Selected counts: 0 valid, –1 range failure, quoted 3 type failure, False explicit bool exclusion.

The zero boundary is already modelled and should be treated as recall; use the remaining individual explanations and later fresh record cases for transfer evidence. Preserve structure versus authority/truth distinction.

Diagnose the current misconceptions

The worked result establishes every future case: **Schema**-valid **input** can still be unauthorised or semantically unsuitable; subsequent checks remain necessary.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Classify counts 0, -1, "3", False under the stated contract.

Teacher explanation: 0 valid; -1 fails range; string fails type; False fails explicit Boolean exclusion.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Typed contracts define values, required fields and failure outcomes. Runtime **input** still needs validation even when program annotations exist.

Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool. Reject extra fields, missing fields and wrong types before **process**ing.

W2 Trace the worked case

Input room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Boolean is an int subclass in Python but excluded by this contract; use type(value) is int and validate nonnegative range.

W3 Audit the limit

A report omits this qualification: **Schema**-valid **input** can still be unauthorised or semantically unsuitable; subsequent checks remain necessary.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: **Schema**-valid **input** can still be unauthorised or semantically unsuitable; subsequent checks remain necessary.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. **Schema**-valid **input** can still be unauthorised or semantically unsuitable; subsequent checks remain necessary.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Boolean is an int subclass in Python but excluded by this contract; use type(value) is int and validate nonnegative range.

Qualification: **Schema**-valid **input** can still be unauthorised or semantically unsuitable; subsequent checks remain necessary.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Classify counts 0, -1, "3", False under the stated contract.

Teacher explanation: 0 valid; -1 fails range; string fails type; False fails explicit Boolean exclusion.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: 0 valid; -1 fails range; string fails type; False fails explicit Boolean exclusion. Require intermediate reasoning and the stated scope. Fresh retry: Counts 5, 5.0, None, True.

Expected reasoning: Only integer 5 passes; float, missing/null and Boolean fail this contract. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled typed contracts and **schema**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of typed contracts and **schema**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Configuration and secrets. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/4, ai-6/6, ai-6/9. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of typed contracts and **schema**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: **Input** room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Boolean is an int subclass in Python but excluded by this contract; use type(value) is int and validate nonnegative range.

Check every learner before discussion: Use the worked case for Typed contracts and **schema**s. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: **Schema**-valid **input** can still be unauthorised or semantically unsuitable; subsequent checks remain necessary.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change A count to -1. Require a **prediction** and a reason before comparing. Expected result: Reject A on the non-negative value rule. Reason: The type is correct but the value is outside its accepted range.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

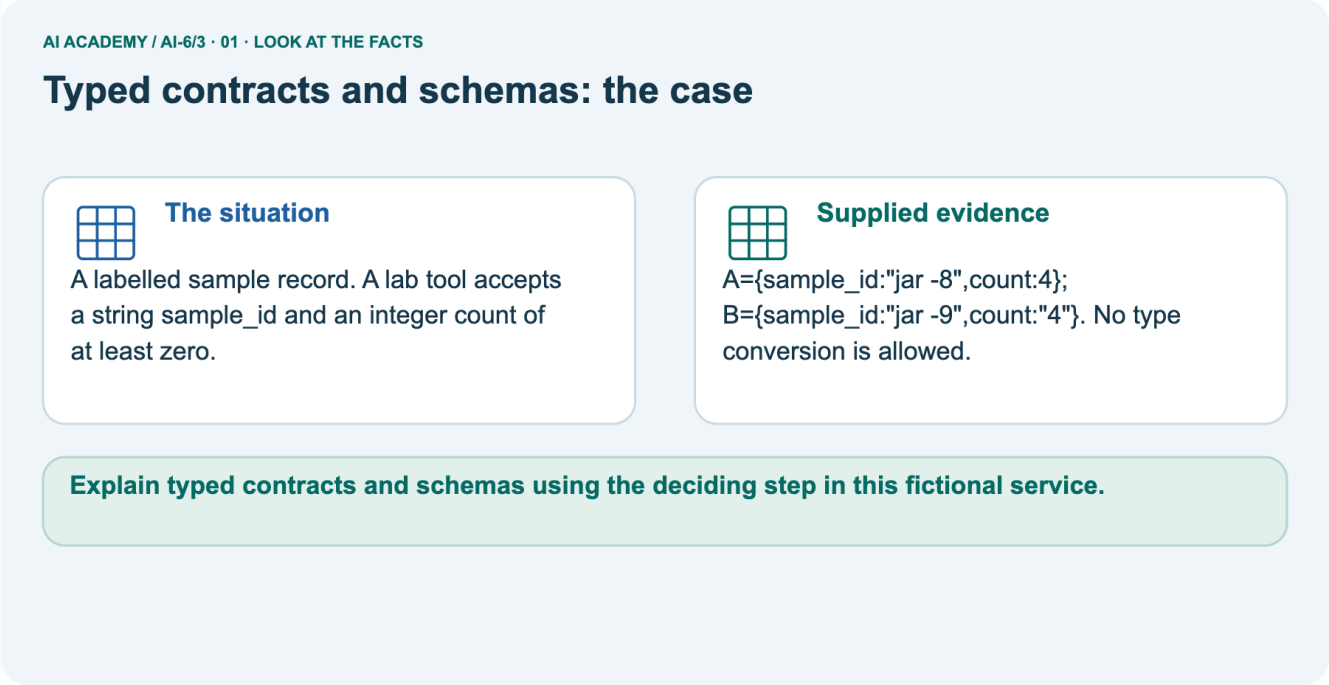
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Typed contracts and schemas: the case



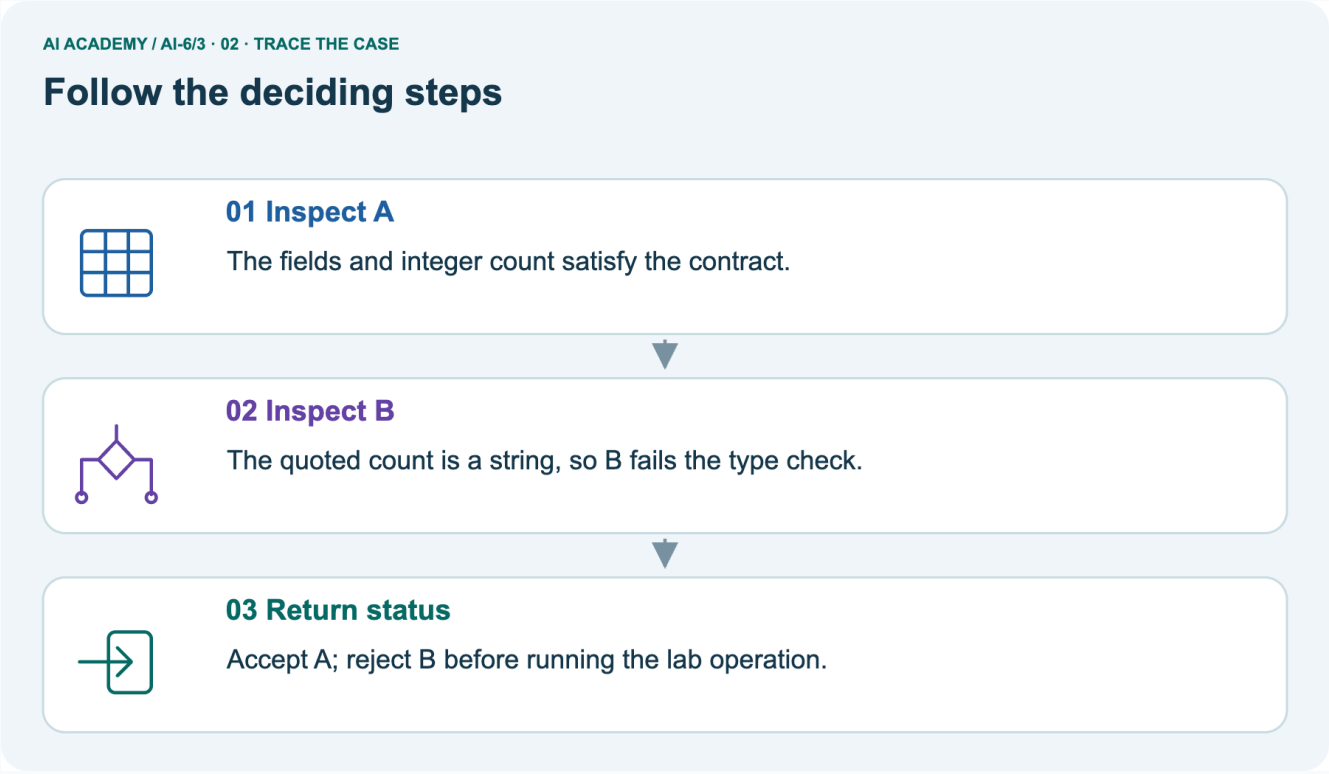
Follow the labelled picture with the matching explanation. Open the vector diagram

A labelled sample record. A lab tool accepts a string sample_id and an integer count of at least zero. Supplied facts: A={sample_id:"jar -8",count:4}; B={sample_id:"jar -9",count:"4"}. No type conversion is allowed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



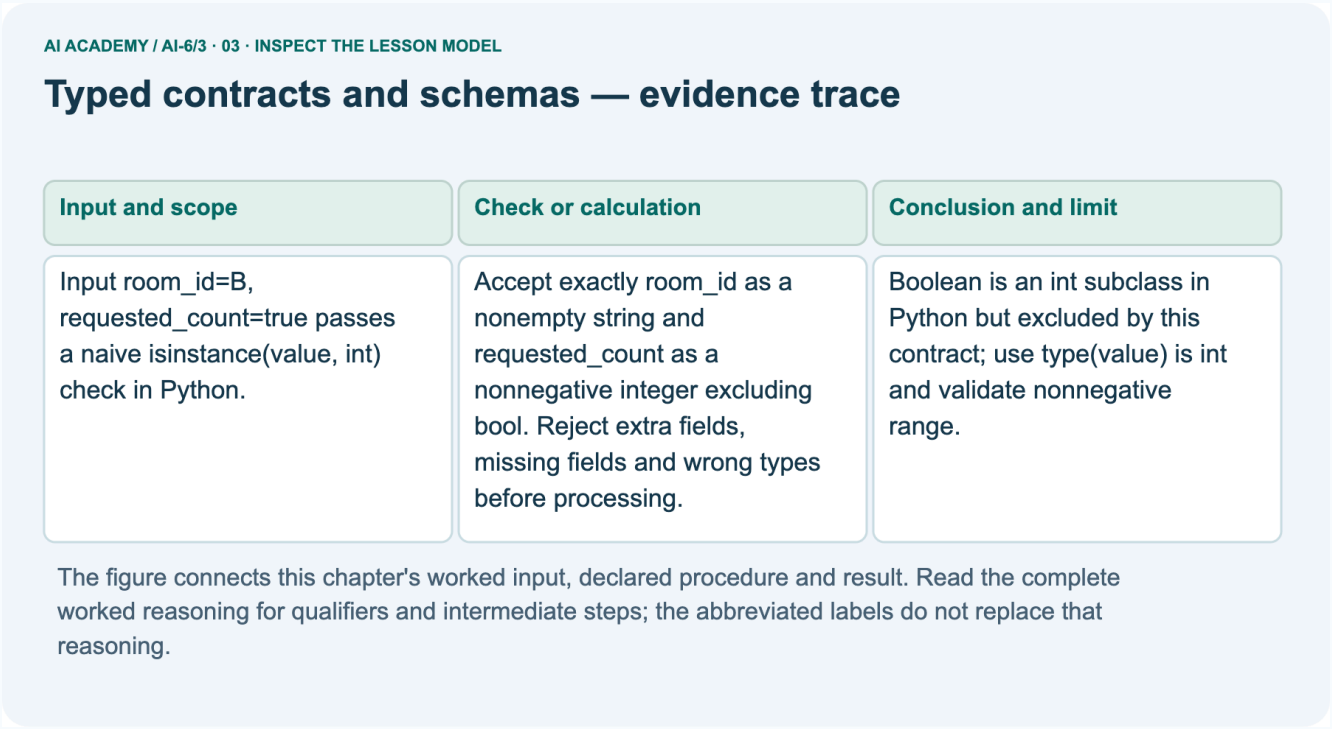
Follow the labelled picture with the matching explanation. Open the vector diagram

Inspect A: The fields and integer count satisfy the contract. → Inspect B: The quoted count is a string, so B fails the type check. → Return status: Accept A; reject B before running the lab operation.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Typed contracts and schemas — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Typed contracts and **schema**s — **evidence** trace.

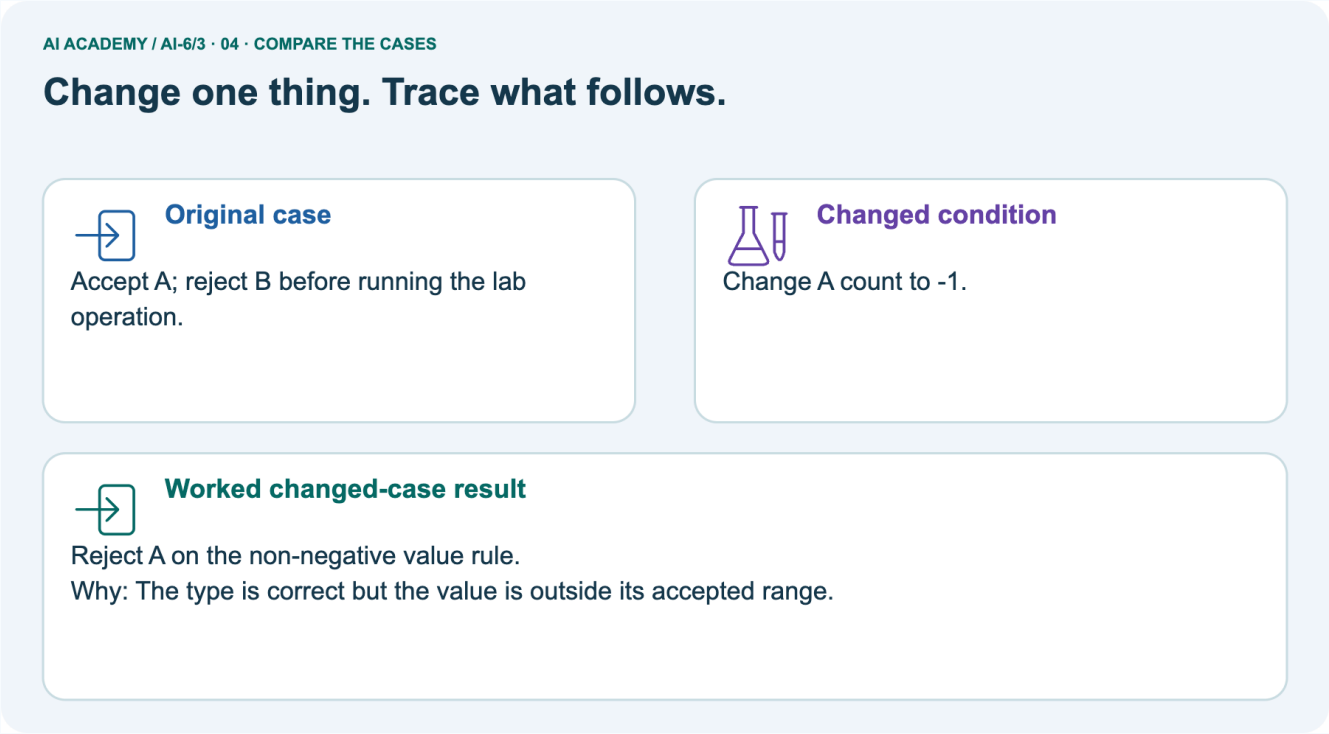
Input room_id=B, requested_count=true passes a naive isinstance(value, int) check in Python. | Accept exactly room_id as a nonempty string and requested_count as a nonnegative integer excluding bool.

Reject extra fields, missing fields and wrong types before **process**ing. | Boolean is an int subclass in Python but excluded by this contract; use type(value) is int and validate nonnegative range.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Accept A; reject B before running the lab operation. Changed condition: Change A count to -1. Expected result: Reject A on the non-negative value rule.

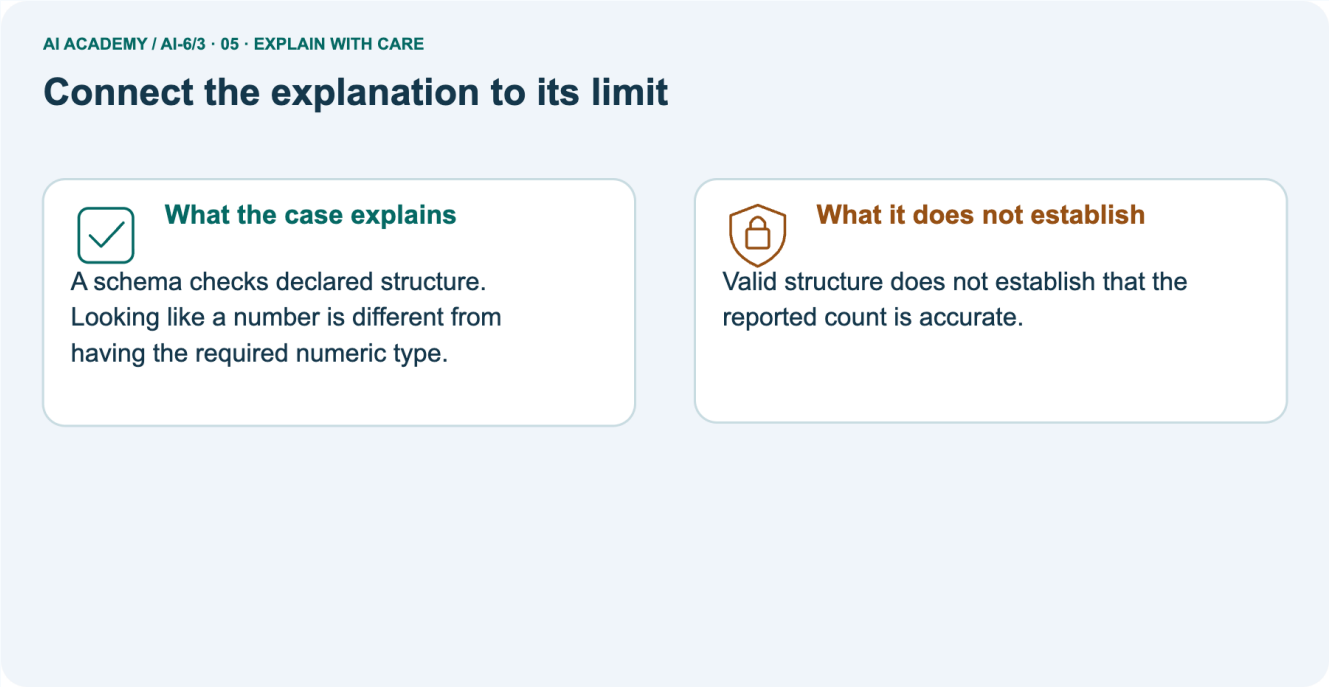
Reason: The type is correct but the value is outside its accepted range.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The type is correct but the value is outside its accepted range.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

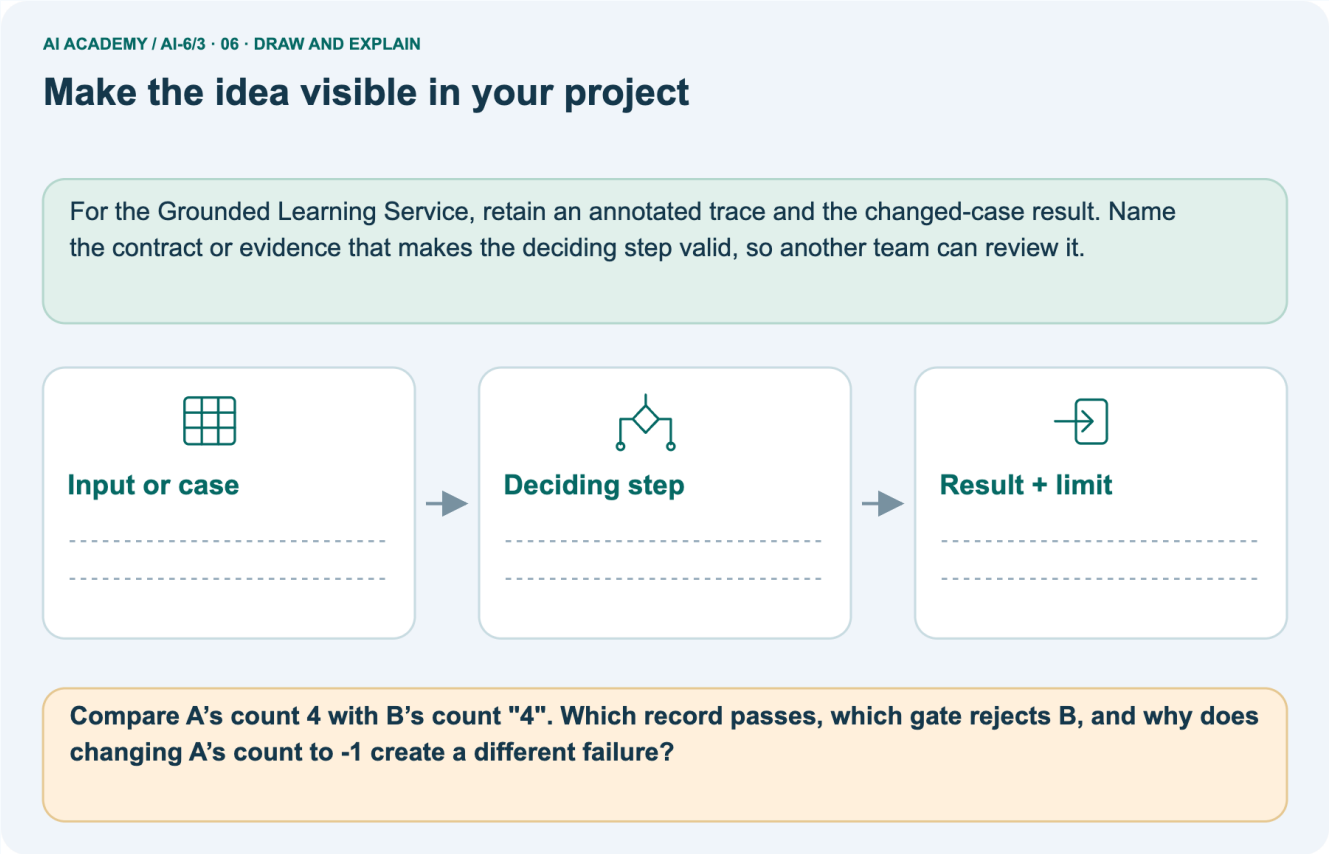
A **schema** checks declared structure. Looking like a number is different from having the required numeric type. Limit: Valid structure does not establish that the reported count is accurate.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Valid structure does not establish that the reported count is accurate.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Compare A's count 4 with B's count "4". Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The fields and integer count satisfy the contract. The quoted count is a string, so B fails the type check. Accept A; reject B before running the lab operation.

Changed case: Reject A on the non-negative value rule. The type is correct but the value is outside its accepted range.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Typed contracts and schemas: the case

02. Follow the deciding steps

03. Typed contracts and schemas — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.